

Real-rootedness of antichain polynomials for $[2] \times [m] \times [n]$

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Abstract

Ding and Dong conjectured that the antichain generating polynomial of the product poset $[2] \times [m] \times [n]$ is real-rooted for all positive integers m, n . Independently, He, Langner, and Witek found by numerical experimentation a 2×2 determinantal formula for the corresponding Zhang–Zhang polynomial, which is obtained from the antichain polynomial by a unit variable shift, but left the formula without a proof. We prove the translated determinantal formula by a weight-preserving tail-switching bijection on pairs of two-rowed arrays. The determinant then factors, after a Möbius change of variables, into two controlled linear combinations of adjacent Jacobi polynomials. Interlacing and endpoint signs show that every finite zero of both factors lies in the required interval. Consequently the antichain generating polynomial of $[2] \times [m] \times [n]$ has only negative real zeros for all positive m, n , proving the conjecture.

1 Introduction

For a finite poset P , let

$$N_P(x) = \sum_{A \in \mathcal{A}(P)} x^{|A|}$$

be its antichain generating polynomial, where the empty antichain is included. Ding and Dong studied these polynomials for products involving minuscule posets and proposed several positivity and real-rootedness conjectures [1]. One of them, [1, Conjecture 4.3], states that

$$N_{[2] \times [m] \times [n]}(x)$$

is real-rooted for every pair of positive integers m, n .

The same family appears in the theory of extended strict order polynomials and Zhang–Zhang polynomials of benzenoid systems. He, Langner, and Witek reported a determinantal expression for the Zhang–Zhang polynomial of the hexagonal flake $O(2, m, n)$ [2]. Their formula was discovered through extensive numerical experimentation and was explicitly stated without proof. The general relation between extended strict order polynomials and the corresponding benzenoid polynomials was developed in [3,4]. If y is the Zhang–Zhang variable and z is

the antichain variable, their interface is $z = y + 1$. We retain x for the antichain variable throughout the proof and make this translation explicit in Section 2.

Our first result proves that determinant directly in the language of posets. Define

$$A_{m,n}(x) = \sum_{j \geq 0} \binom{m}{j} \binom{n}{j} x^j, \quad (1.1)$$

$$B_{m,n}(x) = \sum_{r \geq 0} \binom{m+1}{r+2} \binom{n-1}{r} x^{r+1}, \quad (1.2)$$

and

$$C_{m,n}(x) = \sum_{s \geq 0} \binom{m-1}{s} \binom{n+1}{s+2} x^{s+1}. \quad (1.3)$$

As usual, a binomial coefficient is zero when its lower index lies outside its natural range.

Theorem 1.1. *For all positive integers m, n ,*

$$N_{[2] \times [m] \times [n]}(x) = A_{m,n}(x)^2 - B_{m,n}(x)C_{m,n}(x). \quad (1.4)$$

The proof is a signed tail-switching argument on two-rowed arrays. Its local move is related to the crossing switch used by Krattenthaler [5], but an independent crosscut argument is needed to prove that every negative object has an admissible inverse cut. This avoids the defective global crossing proof corrected by Rubey [6].

Our main result proves the conjecture and directly locates its zeros.

Theorem 1.2. *For all positive integers m, n , every zero of $N_{[2] \times [m] \times [n]}(x)$ is real and negative.*

The proof converts the two factors of (1.4) into linear combinations of adjacent Jacobi polynomials. The coefficients fall exactly in the interval for which the zeros remain in $[-1, 1]$; the Möbius transformation back to x sends the finite zeros to $(-\infty, 0)$.

The paper is organized as follows. Section 2 encodes antichains by two-rowed arrays. Section 3 proves the tail-switching bijection and Theorem 1.1. Section 4 establishes the Jacobi factorization and proves Theorem 1.2. Section 5 records the scope of the result and its relation to the existing literature.

2 Antichains and balanced two-rowed arrays

Write $[r] = \{1, \dots, r\}$ with its natural order. Within one fixed first-coordinate layer of $[2] \times [m] \times [n]$, list the selected elements as

$$(a_1, c_1), \dots, (a_p, c_p),$$

where

$$a_1 < \dots < a_p, \quad c_1 > \dots > c_p.$$

Putting $b_i = n + 1 - c_i$ gives a balanced two-rowed array

$$U = \begin{pmatrix} a_1 < \cdots < a_p \\ b_1 < \cdots < b_p \end{pmatrix}, \quad a_i \in [1, m], \quad b_i \in [1, n]. \quad (2.1)$$

Conversely, every array of the form (2.1) uniquely determines an antichain in $[m] \times [n]$. Hence the length-generating polynomial of one layer is $A_{m,n}(x)$.

Let

$$U = \begin{pmatrix} a_1 < \cdots < a_p \\ b_1 < \cdots < b_p \end{pmatrix}, \quad V = \begin{pmatrix} c_1 < \cdots < c_q \\ d_1 < \cdots < d_q \end{pmatrix} \quad (2.2)$$

encode the two first-coordinate layers. An element from the first layer and an element from the second layer are comparable precisely when some i, j satisfy

$$a_i \leq c_j, \quad b_i \geq d_j. \quad (2.3)$$

Shift the second array by setting

$$c'_j = c_j + 1 \in [2, m + 1], \quad d'_j = d_j - 1 \in [0, n - 1].$$

Then (2.3) is equivalent to the strict crossing condition

$$a_i < c'_j, \quad d'_j < b_i. \quad (2.4)$$

Thus $A_{m,n}(x)^2$ counts all ordered pairs (U, V) , and the antichains of the three-dimensional box are exactly the pairs having no crossing (2.4).

For later comparison, antichains also admit the following standard strict-map interpretation. If P is a finite poset, project an antichain in $[r] \times P$ onto an induced subposet $Q \subseteq P$. Its first coordinates give a strict order-reversing map $Q \rightarrow [r]$, and this correspondence is bijective. Therefore

$$N_{[r] \times P}(x) = E_{P^*}^o(r, x), \quad (2.5)$$

where E^o is the extended strict order polynomial. Since $[2] \times [m]$ is self-dual, (2.5) identifies the polynomial considered here with the relevant extended strict order polynomial. More precisely, using z for the strict-order/antichain variable and y for the Zhang–Zhang variable, the equivalence theorem of Langner and Witek [4] gives

$$\text{ZZ}(O(2, m, n), y) = E_{[2] \times [m]}^o(n, y + 1) = N_{[2] \times [m] \times [n]}(y + 1). \quad (2.6)$$

Thus Theorem 1.1 proves the determinant of [2] after the explicit substitution $x = y + 1$; the two polynomials are not literally identical in the same variable.

3 A two-gap tail-switching bijection

We first describe the forward switch. Give a crossing (i, j) of (2.4) the coordinate

$$S(i, j) = (c'_j, b_i)$$

and order these coordinates lexicographically. Let (I, J) be the crossing with maximal coordinate. Define

$$X_{\text{top}} = (a_1, \dots, a_I, c'_J, \dots, c'_q),$$

$$X_{\text{bot}} = (b_1, \dots, b_{I-1}, d'_{J+1}, \dots, d'_q), \quad (3.1)$$

$$Y_{\text{top}} = (c'_1, \dots, c'_{J-1}, a_{I+1}, \dots, a_p),$$

and

$$Y_{\text{bot}} = (d'_1, \dots, d'_J, b_I, \dots, b_p). \quad (3.2)$$

All four rows are strictly increasing. The two inequalities at the switch are exactly (2.4). If $b_{I-1} \geq d'_{J+1}$, then $(I, J+1)$ would be a crossing with larger first coordinate. If $c'_{J-1} \geq a_{I+1}$, then $(I+1, J)$ would be a crossing with the same first coordinate and larger second coordinate. Empty prefixes and suffixes handle the boundary cases.

The top row of X has two more entries than its bottom row, while the bottom row of Y has two more entries than its top row. The corresponding generating functions are $B_{m,n}(x)$ and $C_{m,n}(x)$. Indeed, if the shorter row of X has length r , then its rows choose $r+2$ entries from $[1, m+1]$ and r entries from $[1, n-1]$ and carry weight x^{r+1} . The analogous statement for Y gives (1.3). The switch preserves the union of the four rows and hence preserves the weight

$$x^{\frac{1}{2}(\text{sum of the four row lengths})}.$$

It remains to prove that the switch is a bijection onto all pairs of arrays of these two unbalanced types.

Lemma 3.1 (two-gap tail switch). *Let*

$$X = \begin{pmatrix} \alpha_1 < \dots < \alpha_{r+2} \\ \beta_1 < \dots < \beta_r \end{pmatrix}, \quad Y = \begin{pmatrix} \gamma_1 < \dots < \gamma_s \\ \delta_1 < \dots < \delta_{s+2} \end{pmatrix}, \quad (3.3)$$

where

$$\alpha \subseteq [1, m+1], \quad \beta \subseteq [1, n-1], \quad \gamma \subseteq [2, m], \quad \delta \subseteq [0, n].$$

Put

$$\gamma_0 = 1, \quad \gamma_{s+1} = m+1, \quad \beta_0 = 0, \quad \beta_{r+1} = n.$$

Then there is a pair $(i, j) \in [1, r+1] \times [1, s+1]$ such that

$$\alpha_i < \gamma_j, \quad \gamma_{j-1} < \alpha_{i+1}, \quad (3.4)$$

$$\beta_{i-1} < \delta_{j+1}, \quad \delta_j < \beta_i. \quad (3.5)$$

Call such a pair an admissible gap. Choose the admissible gap for which

$$T(i, j) = (\alpha_{i+1}, \delta_{j+1})$$

is lexicographically maximal. The reverse splice at this gap produces a unique crossing pair (U, V) , and applying (3.1)–(3.2) at its maximal crossing returns (X, Y) .

Proof. Set

$$\widehat{\beta} = (0, \beta_1, \dots, \beta_r, n), \quad \widehat{\gamma} = (1, \gamma_1, \dots, \gamma_s, m+1).$$

Inside the rectangle $R = [1, m+1] \times [0, n]$, join consecutive points of

$$P_i = (\alpha_i, \widehat{\beta}_i), \quad 1 \leq i \leq r+2,$$

and of

$$Q_j = (\widehat{\gamma}_j, \delta_j), \quad 1 \leq j \leq s+2.$$

Both coordinates increase strictly along every edge. The polyline P joins the bottom side of R to its top side, while Q joins the left side to the right side. The rectangle crosscut lemma therefore implies that they intersect.

Choose incident edges at an intersection so that both their open horizontal projections and their open vertical projections overlap. Such a choice is immediate at an interior intersection. At a common vertex, take the two entering edges or the two leaving edges; strict monotonicity gives the same conclusion. If the polylines share a segment, use an interior point of that segment. At a boundary intersection, the only apparently exceptional entering–leaving choice would force one of the adjacent strictly increasing edges immediately outside R , so it cannot occur. This also covers the four corners.

For the selected edge of P , say the edge from P_i to P_{i+1} , and the selected edge of Q , say the edge from Q_j to Q_{j+1} , overlap of the two open coordinate projections says

$$\alpha_i < \widehat{\gamma}_{j+1}, \quad \widehat{\gamma}_j < \alpha_{i+1},$$

$$\widehat{\beta}_i < \delta_{j+1}, \quad \delta_j < \widehat{\beta}_{i+1}.$$

Substitution of the augmented entries gives (3.4)–(3.5), proving that an admissible gap exists.

At an admissible gap define

$$U_{\text{top}} = (\alpha_1, \dots, \alpha_i, \gamma_j, \dots, \gamma_s),$$

$$U_{\text{bot}} = (\beta_1, \dots, \beta_{i-1}, \delta_{j+1}, \dots, \delta_{s+2}), \tag{3.6}$$

$$V'_{\text{top}} = (\gamma_1, \dots, \gamma_{j-1}, \alpha_{i+1}, \dots, \alpha_{r+2}),$$

and

$$V'_{\text{bot}} = (\delta_1, \dots, \delta_j, \beta_i, \dots, \beta_r). \quad (3.7)$$

The four inequalities (3.4)–(3.5) are precisely the conditions that all rows in (3.6)–(3.7) be strictly increasing. Undoing the shift, $V_{\text{top}} = V'_{\text{top}} - 1$ and $V_{\text{bot}} = V'_{\text{bot}} + 1$, gives balanced arrays in the ranges of (2.2). Moreover $\alpha_i < \alpha_{i+1}$ and $\delta_j < \delta_{j+1}$ become a crossing of U and V' at the splice.

The local switch is the two-rowed-array operation (27) of Krattenthaler [5] under the reindexing

$$M_1 = V' : \quad a_j = c'_j, \quad b_{j-1} = d'_j, \quad M_2 = U : \quad c_i = a_i, \quad d_i = b_i.$$

Under this reindexing its crossing coordinate is both $S = (c'_j, b_i)$ on the balanced side and $T = (\alpha_{i+1}, \delta_{j+1})$ on the unbalanced side. The four tails strictly above and to the right of the splice are unchanged. Hence maximal S is carried to maximal T , and conversely; switching again at that coordinate restores all four rows entry by entry. This proves the asserted inverse relation. Notice that we use only the local switch and maximal-coordinate argument of [5]. The separate crosscut argument above supplies existence on the unbalanced side, so the global crossing proof corrected in [6] is not used.

If $m = 1$ or $n = 1$, one of the interior rows may be empty, but each augmented polyline still has at least one edge. The same crosscut and switch arguments therefore include these boundary cases. $\square$

Proof. The forward construction (3.1)–(3.2) and Lemma 3.1 give a weight-preserving bijection between

1. the crossing pairs counted by $A_{m,n}(x)^2$, and
2. all unbalanced pairs counted by $B_{m,n}(x)C_{m,n}(x)$.

Removing the crossing pairs from all pairs leaves exactly the antichains of $[2] \times [m] \times [n]$. This proves (1.4). $\square$

4 Jacobi factorization and location of the zeros

Because (1.4) is symmetric in m, n , assume $m \leq n$ and put

$$d = n - m, \quad t = \frac{1+x}{1-x}, \quad R = m + n + 1. \quad (4.1)$$

Let

$$G(x) = {}_2F_1(1-m, 1-n; 3; x).$$

Elementary binomial identities applied to (1.2)–(1.3) give

$$B_{m,n}(x)C_{m,n}(x) = \binom{m+1}{2} \binom{n+1}{2} x^2 G(x)^2. \quad (4.2)$$

The hypergeometric representation of Jacobi polynomials [7, Eq. 18.5.8] yields

$$A_{m,n}(x) = (1-x)^m P_m^{(0,d)}(t), \quad (4.3)$$

and

$$G(x) = \frac{(1-x)^{m-1}}{\binom{m+1}{2}} P_{m-1}^{(2,d)}(t). \quad (4.4)$$

Set

$$\rho = \sqrt{\frac{n(n+1)}{m(m+1)}}.$$

Using $x/(1-x) = (t-1)/2$, equations (1.4) and (4.2)–(4.4) give

$$N_{[2] \times [m] \times [n]}(x) = (1-x)^{2m} F_-(t) F_+(t), \quad (4.5)$$

where

$$F_{\pm}(t) = P_m^{(0,d)}(t) \pm \frac{\rho}{2}(t-1)P_{m-1}^{(2,d)}(t). \quad (4.6)$$

Write $Q_k(t) = P_k^{(1,d)}(t)$. The adjacent-parameter identities for Jacobi polynomials [7, Eqs. 18.9.5–18.9.6], after the symmetry $t \mapsto -t$ and exchange of parameters, give

$$P_m^{(0,d)}(t) = \frac{n+1}{R}Q_m(t) - \frac{n}{R}Q_{m-1}(t), \quad (4.7)$$

and

$$(t-1)P_{m-1}^{(2,d)}(t) = \frac{2m}{R}Q_m(t) - \frac{2(m+1)}{R}Q_{m-1}(t). \quad (4.8)$$

Consequently

$$RF_-(t) = (n+1-\rho m)Q_m(t) + (\rho(m+1)-n)Q_{m-1}(t), \quad (4.9)$$

and

$$RF_+(t) = (n+1+\rho m)Q_m(t) - (n+\rho(m+1))Q_{m-1}(t). \quad (4.10)$$

We use the following elementary consequence of interlacing.

Lemma 4.1. *Let $d \geq 0$ and $Q_k = P_k^{(1,d)}$. If*

$$-\frac{m+1}{m} < c < \frac{m+d}{m}, \quad (4.11)$$

then all m zeros of $Q_m + cQ_{m-1}$ lie in $(-1, 1)$. At either endpoint of (4.11), the corresponding endpoint 1 or -1 is a zero and all remaining zeros lie in $(-1, 1)$.

Proof. The Jacobi weight $(1-t)(1+t)^d$ is positive on $(-1, 1)$, so the zeros of Q_m and Q_{m-1} are simple and strictly interlace [7, Sec. 18.2(vi)]. Evaluating $Q_m + cQ_{m-1}$ at consecutive zeros of Q_m gives $m-1$ zeros, one in each intervening interval. The endpoint values are

$$Q_k(1) = k+1, \quad Q_k(-1) = (-1)^k \binom{k+d}{k}.$$

The endpoint values give the bounds

$$c > -\frac{m+1}{m} \quad \text{and} \quad c < \frac{m+d}{m}.$$

If $c > 0$, the $m-1$ interior sign changes are supplemented on the left: the condition $c < (m+d)/m$ makes the signs at -1 and at the smallest zero of Q_m opposite. If $c < 0$, the condition $c > -(m+1)/m$ similarly supplies the last zero in the right outer interval. If $c = 0$, the polynomial is Q_m itself. Equality in the upper or lower bound makes -1 or 1 , respectively, the final zero. $\square$

We now apply the lemma. The coefficient ratio in (4.9) is

$$c_- = \frac{\rho(m+1) - n}{n+1 - \rho m}. \quad (4.12)$$

Both numerator and denominator are positive, as follows explicitly from

$$\rho^2(m+1)^2 - n^2 = \frac{n(m+n+1)}{m} > 0$$

and

$$(n+1)^2 - \rho^2 m^2 = \frac{(n+1)(m+n+1)}{m+1} > 0.$$

Moreover

$$c_- \leq \frac{n}{m},$$

because this inequality reduces to $\rho m \leq n$, whose square is equivalent to $m \leq n$. Equality holds exactly when $m = n$. Thus all zeros of F_- lie in $(-1, 1)$ when $m < n$; when $m = n$, one zero is $t = -1$ and the others remain in $(-1, 1)$.

The coefficient ratio in (4.10) is

$$c_+ = -\frac{n + \rho(m+1)}{n+1 + \rho m}. \quad (4.13)$$

Clearly $c_+ < n/m$, while

$$c_+ > -\frac{m+1}{m}$$

reduces, after cancellation of the common $\rho m(m+1)$ term, to $mn < (m+1)(n+1)$. Hence every zero of F_+ lies in $(-1, 1)$.

Proof. The inverse transformation in (4.1) is

$$x = \frac{t-1}{t+1},$$

which maps $(-1, 1)$ bijectively onto $(-\infty, 0)$. Equations (4.5) and the preceding analysis therefore place every finite zero of $N_{[2] \times [m] \times [n]}(x)$ on the negative real axis. When $m = n$, the endpoint zero $t = -1$ of F_- corresponds to the point at infinity and accounts for the degree drop from $2m$ to $2m - 1$; it creates no finite zero. This proves the theorem. $\square$

5 Consequences, scope, and prior-work boundary

Real-rootedness immediately implies log-concavity and unimodality of the coefficient sequence because all coefficients are nonnegative. Theorem 1.2 also records the zero location explicitly: every zero lies in $(-\infty, 0)$. For a real-rooted polynomial with positive coefficients and constant term 1, this location is a consequence of real-rootedness rather than a logically stronger claim.

After the substitution $x = y + 1$ from (2.6), Theorem 1.1 proves the determinant of [2]. It also explains the shifted parameters in its off-diagonal entries: they arise from the two-unit row imbalance forced by the tail switch rather than from numerical fitting. The proof is entirely within poset and two-rowed-array combinatorics; the benzenoid interpretation is not needed.

The argument is specific to the two-layer factor [2]. For $[k] \times [m] \times [n]$ with $k > 2$, one would need a higher-order determinant or a compatible multi-path cancellation. No such extension is claimed here. We also do not address the separate γ -positivity conjecture for $N_{[2] \times [n] \times [n+1]}(x)$ from [1].

As of 26 August 2026, targeted searches of the exact conjecture, the determinant in [2], the $O(2, m, n)$ family, and the extended strict order-polynomial literature found no public proof of either Theorem 1.1 or Theorem 1.2. This records the search boundary, not a claim of exhaustive worldwide priority.

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References

- [1] J. Ding and C.-P. Dong, *Antichain generating polynomials of posets*, arXiv:1905.06692v1 (2019).
[arXiv:1905.06692](https://arxiv.org/abs/1905.06692).

- [2] B.-H. He, J. Langner, and H. A. Witek, *Hexagonal flakes as fused parallelograms: A determinantal formula for Zhang–Zhang polynomials of the $O(2, m, n)$ benzenoids*, J. Chinese Chem. Soc. 68 (2021), 1231–1242. [doi:10.1002/jccs.202000420](https://doi.org/10.1002/jccs.202000420).
- [3] J. Langner and H. A. Witek, *Extended strict order polynomial of a poset and fixed elements of linear extensions*, Australas. J. Combin. 81 (2021), 187–207. [Open-access article](#).
- [4] J. Langner and H. A. Witek, *ZZ polynomials of regular m -tier benzenoid strips as extended strict order polynomials of associated posets—Part 1: proof of equivalence*, MATCH Commun. Math. Comput. Chem. 87 (2022), 585–620. [doi:10.46793/match.87-3.585L](https://doi.org/10.46793/match.87-3.585L).
- [5] C. Krattenthaler, *Counting nonintersecting lattice paths with turns*, Sémin. Lothar. Combin. 34 (1995), Article B34i, 17 pp. [Open-access article](#).
- [6] M. Rubey, *Comment on “Counting nonintersecting lattice paths with turns” by C. Krattenthaler*. [Open-access comment](#).
- [7] NIST Digital Library of Mathematical Functions, Chapter 18: *Orthogonal Polynomials*. dlmf.nist.gov/18.